Glaucoma classification using a morphological-convolutional neural network trained with extreme learning machine

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ABSTRACT

Mathematical morphology is a technique frequently used in image processing, it has several applications such as segmentation, filtering, compression, edge detection, and feature extraction. Considering this last application here is presented a morphological and convolutional neural network (MCNN) that takes advantage of the different types of morphological operations – erosion, dilation, opening, and closing – by including them in a single layer. Three independent neural networks are used to learn information per channel, Random Forest is used at the end, fed with the three outputs of the NNs. The classification performance of the method was compared against three common CNN architectures: ResNet-18, ShuffleNet-V2, and MobileNet-V2. Two training approaches were used: training from scratch and using transfer learning. A glaucoma classification was conducted using the ORIGA dataset. The MCNN method obtained an AUC of 0.704 (0.672, 0.743 95% CI) with a performance similar to the other CNN methods when trained using transfer learning.

Keywords: medical image classification, mathematical morphology, glaucoma

1. INTRODUCTION

Medical images have been extensively used to support diagnosis, traditionally the identification of the illness is a job performed by specialized physicians, however, it implies high consultation charges and imprecise interpretations resulting from natural human error. To overcome these difficulties, artificial intelligence (AI) has been proposed, we can find two popular approaches: hand-craft features coupled with traditional machine learning and convolutional neural networks (CNN). The former approach has provided satisfactory results, however, the feature extraction process is time-consuming and requires high effort. Contrary, CNN learns the features needed and extracts them on its own.[1] Several architectures of CNNs have been proposed until now, most of them contain a high number of layers and parameters. Architectures such as ResNet[2], Inception networks[3, 4], MobileNet[5], ShuffleNet[6], and DenseNet[7] have been widely adopted and used in medical imaging applications for COVID-19 diagnosis at chest CT[8, 9], classification of fibrotic lung disease[10, 11], classification of skin cancer[12], and detection of acute intracranial hemorrhage[13].

Although CNN has been successful in several medical image classification applications, it presents some disadvantages: hyper-parameters optimization (learning rate, optimizer, dropout rate, etc.) plays a determinant role in the performance, it can help to improve it or decline it; to capture low-level textural information in the images, small-sized kernels are preferable, however, this increase the computational complexity during training; CNN with significant depth and enormous size require huge memory and higher computational resources for training, due to the involvement of millions of training parameters which could lead the model to overfit and be inefficient at generalization, especially in the case of limited dataset, this calls for models which are lightweight and which could also extract critical features like the dense models[1]; it requires an excessive amount of images since insufficient information or features can lead to underfitting the model, alternative tools have been proposed to overcome this problem, such as data augmentation[14] or transfer learning[15]. However, despite the high performance of many medical imaging models pretrained with non-medical imaging datasets, previous works[16-18] have shown that pretrained models developed from medical source databases could achieve better performance, successful transfer learning requires a reasonably large sample size, diversity of images, and similarity between the training and the target application images[19]. Regarding data

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augmentation, it is important to consider that it's a technique that requires additional memory and computation costs to conduct the transformations, besides, it is required more effort, first for selecting the appropriate transformation for each specific case, and second, for manually observing to make sure they have not altered the label of the image or the information that would be relevant. Additionally, the biases may distance the training from the testing data[14].

Regarding the limited access to datasets large enough to handle AI, high-quality images and annotations are still essential for supervised training, validation, and testing of commercial AI algorithms. This is especially true in the clinical setting and is well outlined by Park and Han[20]. Small sample sizes and a lack of diverse geographic areas hinder the generalizability and accuracy of developed solutions [21]. Most research groups and industries dedicated to AI medical methods development, struggle with the available supply of medical image datasets for research purposes because they are in scarcity and image acquisition and preparation is a process with a lot of difficulties. First, it is costly and time-intensive: it is needed access to appropriate clinical installations with expensive medical devices; adequate curation, anonymization, analysis, or annotations of clinical data implies high difficulty[22]; regarding image labeling, it needs to be performed by medical experts such as radiologists [23]. Second, even when datasets are open source, an additional human inspection of each image is required because some images contain free-form annotations that have been scanned and cannot be removed reliably with automated methods [23]. Only a few well-curated annotated medical imaging datasets with high-quality ground truth pathologic labels are publicly available[24]. The U.K. Biobank[25, 26] contains multimodal images in more than 100000 participants; however, most U.K. participants are healthy. The Cancer Imaging Archive[27] contains a large database of national lung screening trials, which consists of low-dose helical CT images in 53454 participants[28]. The National Institutes of Health ChestX-ray8 contains 108948 chest radiographs in 32717 patients, with eight disease labels[29].

All the previously mentioned complications can be reduced if deep learning researchers include into their priorities the development of methods able to achieve good performance with small datasets. Therefore, here we are proposing a morphological-convolutional neural network (MCNN) for medical image classification trained with extreme learning machine (ELM) and combining the potential of Random Forest (RF), that avoids a deep structure that requires a high amount of data to be appropriately trained, it consists of a compacted design that considers the capability of mathematical morphology for the identification of important descriptors in specific medical cases. We evaluated the performance of the method for glaucoma classification. This article has been organized as follows: Section 2 presents the methodology used, Section 3 shows the results obtained, Section 4 includes the discussion of our findings, and finally, we present our conclusion in Section 5.

2. METHODOLOGY

2.1 Morphological-convolutional neural network

In this method, we aimed to combine the capabilities of both, convolutional and morphological operations. We used a CNN-like architecture with an additional layer performing different types of morphological operations.

2.1.1 Morphological layer

We included erosion, dilation, opening, closing, and morphological residual – the erosion is subtracted from the original image – for grayscale images.

We included these operations in a single layer where they are parallelly applied. To create the opening and closing operations we added a sequence operation – which permits to apply the operations sequentially, resulting in the creation of openings and closings – in the layer. A different filter is learned per operation. In Figure 1c we find a graphical description of the internal configuration of the morphological layer.

2.1.2 Three-channels approach

As the morphological operations in the layer work only for grayscale images, we included three neural networks for independent training whose output probabilities are used as inputs of a Random Forest (RF) classifier[30]. Each of these neural networks focuses on the feature extraction of a specific color channel. Each neural network is trained using ELM[31]. The MCNN method with the three channels approach is shown in Figure 1a.

2.1.3 Complete architecture

The final architecture is summarized in Figure 1. The operations used in the morphological layer for this study were dilation, erosion, opening, and residual morphological where the erosion was extracted from the original image. One morphological layer, one convolutional and a fully connected were implemented per channel (Figure 1b).

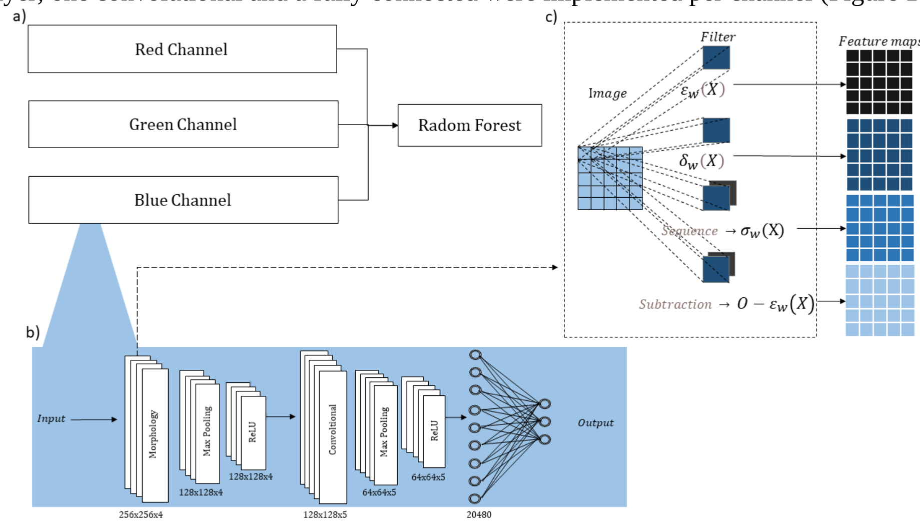

Figure 1. Architecture overview: a) The probability outputs (a neural network per channel) are the inputs of a Random Forest classifier; b) Detailed architecture of each neural network; c) inside the morphological layer, four operations are parallelly applied.

2.2 Dataset

To evaluate the performance of the proposed method for glaucoma classification, we expect adequate results since for glaucoma identification the shape of the blood vessels near the optic disk is a key factor.

The dataset considered corresponds to the Online Retinal Fundus Image Database for Glaucoma Analysis and Research (ORIGA-light)[32] (Figure 2a and 2b). The dataset contains 650 fundus images, 168 for glaucoma and 482 for non-glaucoma. Each image was annotated by trained professionals. The original size of the images is 2048x3072, we resized them to 256x256 pixels.

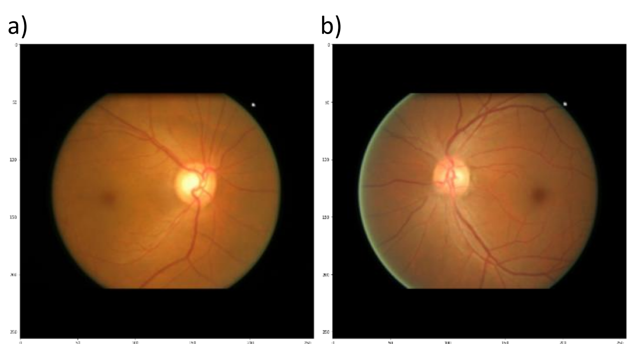

Figure 2. Main datasets for the study: a) fundus image corresponding to glaucoma; b) fundus image corresponding to a non-glaucoma eye disease.

Additionally, we used a dataset of easy classification to ensure our settings were appropriate: the German Traffic Sign Recognition Benchmark (GTSRB)[33], it is a multi-class classification problem (43 classes), however, considering the scope of our study we limited it to two classes (1 and 13). The images come in a huge variety of sizes, for avoiding drastic image distortion, we chose images with a width of fewer than 150 pixels and resized them to 100 pixels.

2.3 Classification specifications

For the classification tasks using the MCNN method, we divided the datasets into 70% for training and 30% for testing, using a stratified split. In this case, as the optimizer that we are using is a classic ELM, a validation set is unrequired. The weights were randomly initialized using a Gaussian distribution and then passed to a threshold around zero to binarize them. In ELM the only parameter requiring tuning is a regularization parameter, we set it to 1×10^{-3}. We executed the method 10 times with different seeds but keeping the same test set and proceed to obtain the median of each classification metrics.

Three common CNN architectures were used for comparison: ResNet-18, ShuffleNet-V2, and MobileNet-V2. We opted for architectures with the smaller number of parameters possible. For these three models, we proceeded to divide into 70%, 10%, and 20% for training, validation, and testing sets, respectively, again we used a stratified split. The loss function used was the binary cross entropy loss, with stochastic gradient descent as optimizer using a learning rate of 0.01, and momentum of 0.9, we used an exponential learning rate scheduler with a gamma of 0.9, and 24 epochs. We followed two approaches with these models: training from scratch, and training using transfer learning. For the transfer learning approach, we used models pre-trained with ImageNet[34], we froze the weights for all of the network except that of the final fully connected layer. This last fully connected layer was replaced with a new one with random weights and only this layer was trained.

Besides the classification performed with the ORIGA datasets, we proceeded to execute an additional benchmark classification with the GTSRB dataset to ensure the settings in the CNN common architectures were suitable.

The resizing of the images and the classification tasks were performed using the Pytorch framework[35] with Python (Python Software Foundation. Python Language Reference, version 3.7.14. Available at http://www.python.org)[36]. A summary of the methodology followed in this study is found in Figure 3.

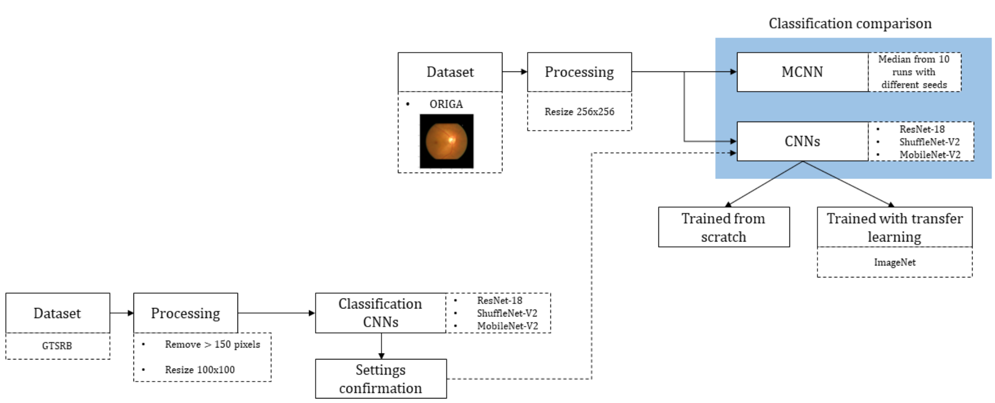

Figure 3. Overview methodology. A benchmark classification is performed using the GTRSB dataset to ensure appropriate setting for CNN architectures. Two main classification tasks – glaucoma vs non-glaucoma and melanoma vs benign – are executed with the MCNN method and common CNN architectures.

3. RESULTS

3.1 MCNN variability

The resulting variability of our model is shown in Figures 4. The median AUC found for the ORIGA dataset corresponded to 0.704 (0.672 – 0.743, 95% CI). A more detailed inspection of the distribution is found in Figure 5 and Table 1.

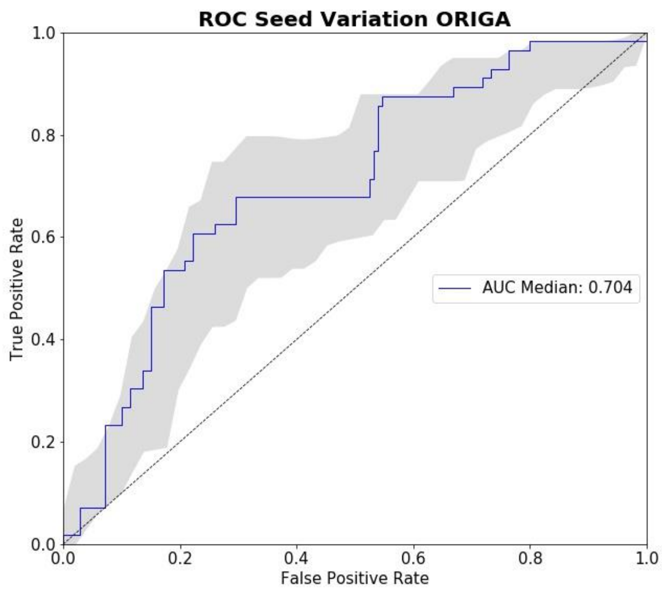

Figure 4. ROC-AUC for glaucoma classification. The method was tested with 10 different seeds and the variability is presented with the shaded part. The median is highlighted with a blue line.

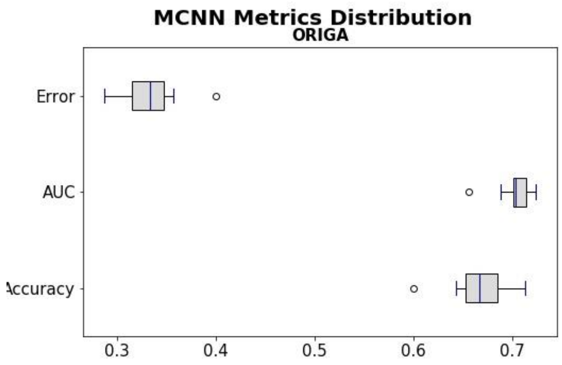

Figure 5. Error, AUC, and accuracy distributions when testing the MCNN method with 10 different seeds, for both, glaucoma (left) and melanoma (right) classification.

Table 1. Statistical metrics for variation of MCNN through runs with different seeds with ORIGA dataset.

	Accuracy	**Balanced Accuracy**	**AUC**	**Error**
Mean	0.6673	0.6581	0.7026	0.3327
Variance	0.0010	0.0009	0.0004	0.0010

Std	0.0322	0.0302	0.0197	0.0322
Median	0.6670	0.6640	0.7035	0.3330

3.2 CNNs benchmark

The results of the benchmark classification with the GTSRB dataset to ensure suitable settings for the CNN architectures are shown in Figure 6. Validation loss and accuracy, as well as the AUC for the three methods – ResNet-18, ShuffleNet-V2, and MobileNet-V2 – are presented. The three methods appropriately converged and reached accuracies of 0.9931, 0.9597, and 0.997 for ResNet-18, ShuffleNet-V2, and MobileNet-V2, respectively.

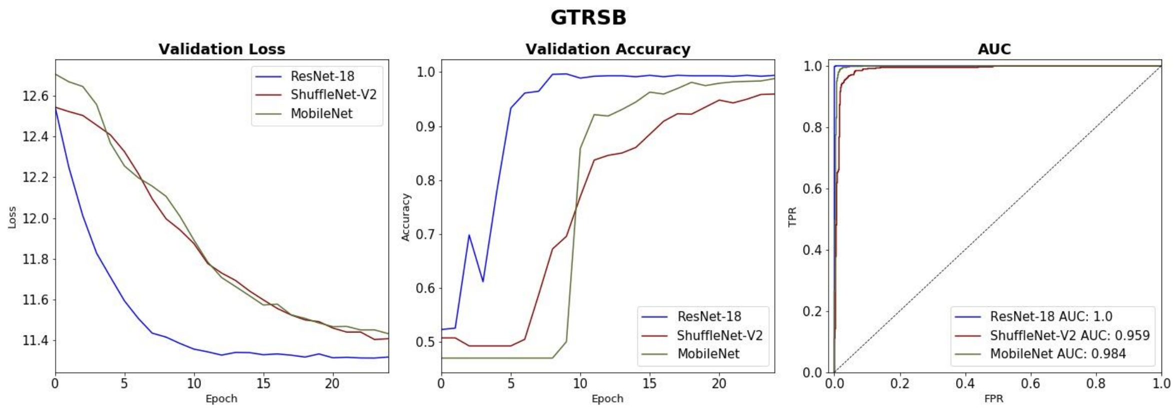

Figure 6. Metrics for traffic signs classification used as benchmark to ensure appropriate settings for the three architectures used for comparison: ResNet-18, ShuffleNet-V2, and MobileNet-V2.

After proving the methods work appropriately by using an easy classification with the GRSB dataset, we continue to explore the performance for the glaucoma classification with the ORIGA dataset. In Figures 7, we present validation losses and accuracies for the three methods when trained from scratch and when using transfer learning.

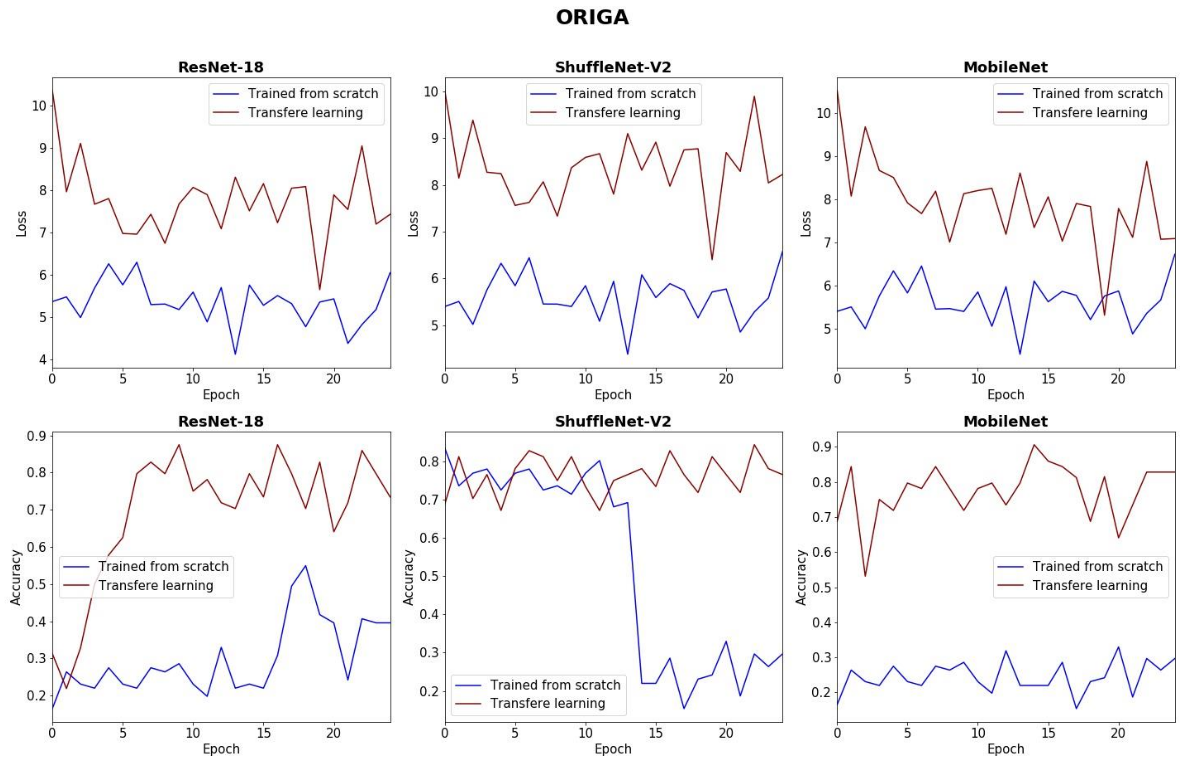

Figure 7. Validation loss and accuracy of the three CNN methods when classifying glaucoma. Training from scratch is compared to transfer learning.

3.3 Comparison

We present a comparison between our MCNN method and the three chosen CNN architectures. In Figure 8 we find the ROCs, including both processes: trained from scratch and with transfer learning, and the complete metrics – balanced accuracy, error, AUC, and number of parameters – are presented in Tables 2

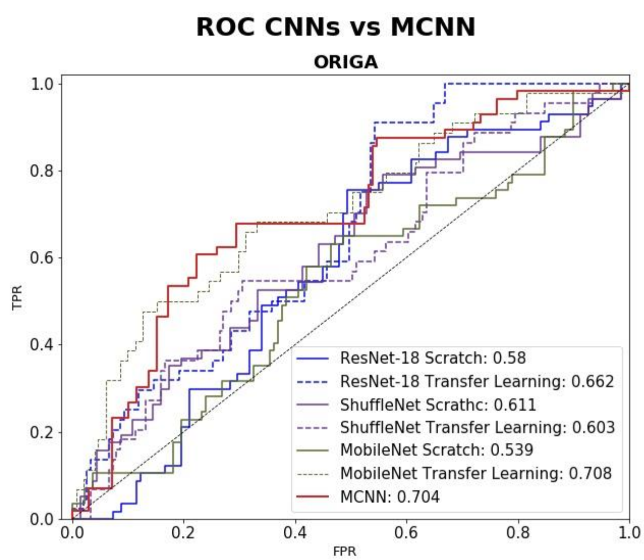

Figure 8. ROC-AUCs for glaucoma classification. Comparison of MCNN model against to ResNet-18, ShuffleNet-V2, and MobileNet-V2, trained from scratch and using transfer learning.

Table 2. Results of glaucoma classification using the ORIGA dataset and number of parameters trained per method.

Model	Accuracy	Balanced Accuracy	AUC	Error	No. of Params
	95% CI	95% CI	95% CI	95% CI	
ResNet-18 Scratch	0.297 (0.233, 0.362)	0.493 (0.423, 0.563)	0.580 (0.496, 0.664)	0.703 (0.638, 0.767)	11177025
ShuffleNet-V2 Scratch	0.651 (0.637, 0.665)	0.589 (0.520, 0.658)	0.611 (0.521, 0.700)	0.023 (0.002, 0.044)	61101841
MobileNet-V2 Scratch	0.292 (0.228, 0.356)	0.500 (0.430, 0.570)	0.539 (0.450, 0.629)	0.708 (0.644, 0.772)	132864337
ResNet-18 Transfer Learning	0.764 (0.705, 0.824)	0.509 (0.439, 0.580)	0.662 (0.580, 0.744)	0.236 (0.176, 0.295)	513
ShuffleNet-V2 Transfer Learning	0.738 (0.677, 0.800)	0.517 (0.447, 0.587)	0.603 (0.508, 0.699)	0.262 (0.200, 0.323)	1025
MobileNet Transfer Learning	0.785 (0.727, 0.842)	0.531 (0.461, 0.601)	0.708 (0.618, 0.798)	0.215 (0.158, 0.273)	1001
MCNN	0.667 (0.628, 0.711)	0.664 (0.621, 0.705)	0.704 (0.672, 0.743)	0.333 (0.273, 0.390)	61440

4. DISCUSSION

By testing our model with different seeds we found that the distribution of different metrics (accuracy, AUC, and error) has a median of 0.667 for accuracy, 0.704 for AUC and 0.333 error (Table 1). However, the results presented one outlier (Figure 5). Even considering this outlier, we can observe how our model is stable through different seeds with a variance in accuracy of 0.001 (Table 1).

Regarding the performance of the CNN typical architectures – ResNet-18, ShuffleNet-V2, and MobileNet-V2 – none of their validation losses were able to converge when trained with the ORIGA dataset (Figure 7), neither when trained from scratch, nor when trained using transfer learning. The validation accuracy of all the methods obtained a significant improvement when using transfer learning. The same configuration was used with the GTRSB dataset, whose results showed an appropriate convergence behavior and an improvement of validation accuracy through epochs (Figure 6). Considering this, we can conclude that the poor performances rely mostly on the small size and low resolution of the dataset and not on the configuration of the models.

By comparing the AUC results of our model and the three CNN architectures (Figure 8) we observed that our method obtained the second-best performance after MobileNet-V2 trained with transfer learning, considering the value estimated but without a significative difference, our model obtained 0.704 (0.672, 0.743 95% CI) and MobileNet-V2 obtained 0.708 (0.618, 0.798 95% CI). Using this information, we can observe how our method tends to obtain an AUC result similar to the CNN methods mainly when trained with transfer learning.

Among the three CNN architectures, only MobileNet-V2 showed an improvement when trained with transfer learning compared to when trained from scratch, 0.708 (0.618, 0.798 95% CI) against 0.539 (0.45, 0.629 95% CI), respectively. Considering this, we see that does not exist a clear significative difference between the CNN methods when trained from scratch and when trained with transfer learning from the perspective of the AUC result.

Contrary to the AUC result, by considering the accuracy we observe a clear difference between the three CNN methods when trained from scratch and when trained using transfer learning. The three methods with transfer learning obtained the best accuracy without significative difference between them: 0.764 (0.705, 0.824 95% CI), 0.738 (0.677, 0.8 95% CI), 0.785 (0.727, 0.842 95% CI) for ResNet-18, ShuffleNet-V2, and MobileNet-V2, respectively. Both, ResNet-18 and MobileNet-V2 trained from scratch obtained accuracies under 0.37, they are not able to classify. Our method obtained an accuracy lower than the three CNN methods trained with transfer learning (0.667 [0.628, 0.711 95% CI]), however, contrary to some of the methods trained from scratch, it is able to classify, the accuracy of our method is significatively similar to ShuffleNet-V2 (0.651 [0.637, 0.665] 95% CI).

Another observation is that our method responds better to an imbalanced dataset than the other three CNN methods, we have an accuracy of 0.667 (0.628, 0.711 95% CI) and a balanced accuracy of 0.664 (0.621, 0.705), meaning that we have a difference of 0.003 even if the ORIGA dataset is a ~ 74%-26% imbalanced dataset. The rest of the methods have a difference between their accuracies and balanced accuracies of 0.062-0.255.

The MCNN tends to obtain a better performance considering the balanced accuracy. It is better than two of the three CNN methods and has a similar result to ShuffleNet-V2 trained from scratch, 0.664 (0.621, 0.705 95% CI) against 0.598 (0.520, 0.658 95% CI).

5. CONCLUSION

In this study, we are proposing a morphological-convolutional neural network (MCNN) for medical image classification trained with extreme learning machine (ELM) and combining the potential of Random Forest (RF), that avoids a deep structure that requires a high amount of data to be appropriately trained. We evaluated the method for glaucoma classification. We compared our MCNN method against CNN architectures typically found in the literature: ResNet-18, ShuffleNet-V2, and MobileNet-V2, using two training approaches, from scratch and using transfer learning. We found that our MCNN method tends to obtain better results for classification when using a small dataset than typical CNN architectures when trained from scratch and with comparable results to the methods trained using transfer learning.

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